



AIDOS LAB
AI FOR DATA-ORIENTED SCIENCE



Learning on topological data: over-squashing and over-smoothing

Lecture 4 — Geometry and Topology in Machine Learning

Lecture 4 — Learning on Topological data

Outline

- Revisiting message passing: receptive fields
- Partitioning a graph: conductances and the normalized Laplacian
- The spectral method and Cheeger inequalities
- Limitations to message-passing
 - Oversmoothing
 - Oversqueashing

Message-passing revisited

Session 3 — Learning on Topological Data

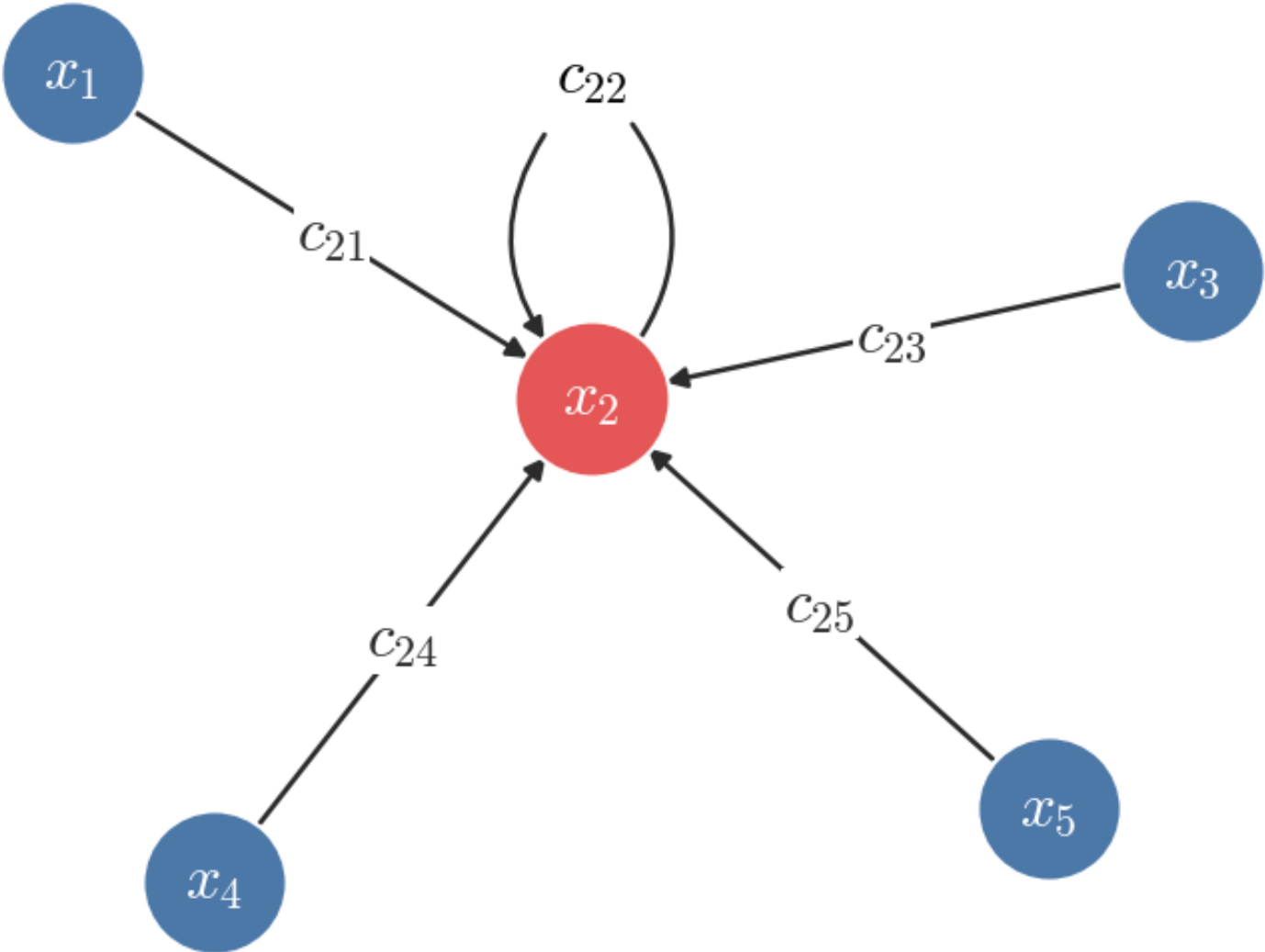
Recap: message-passing

A way of instantiating local functions in GNNs

Initial features $X = \begin{pmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix} \in \mathbb{R}^{n \times k}$

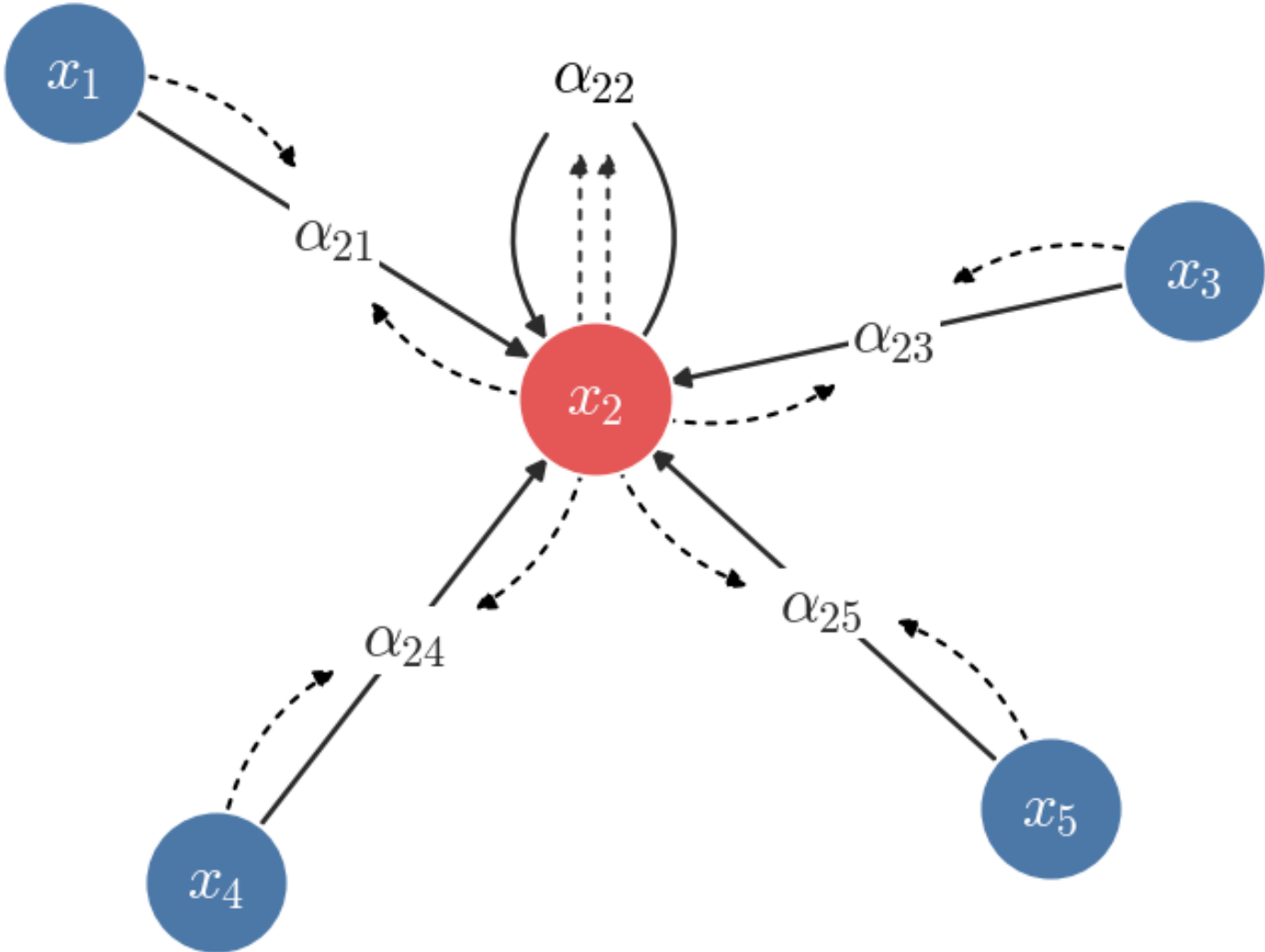
We defined GNNs as equivariant functions $f_\theta(X, A)$ defined from local functions invariant wrt the labelling on the neighbors

One way of achieving such invariance is via *message-passing*, based on aggregating the features over the neighborhoods (the **message**)



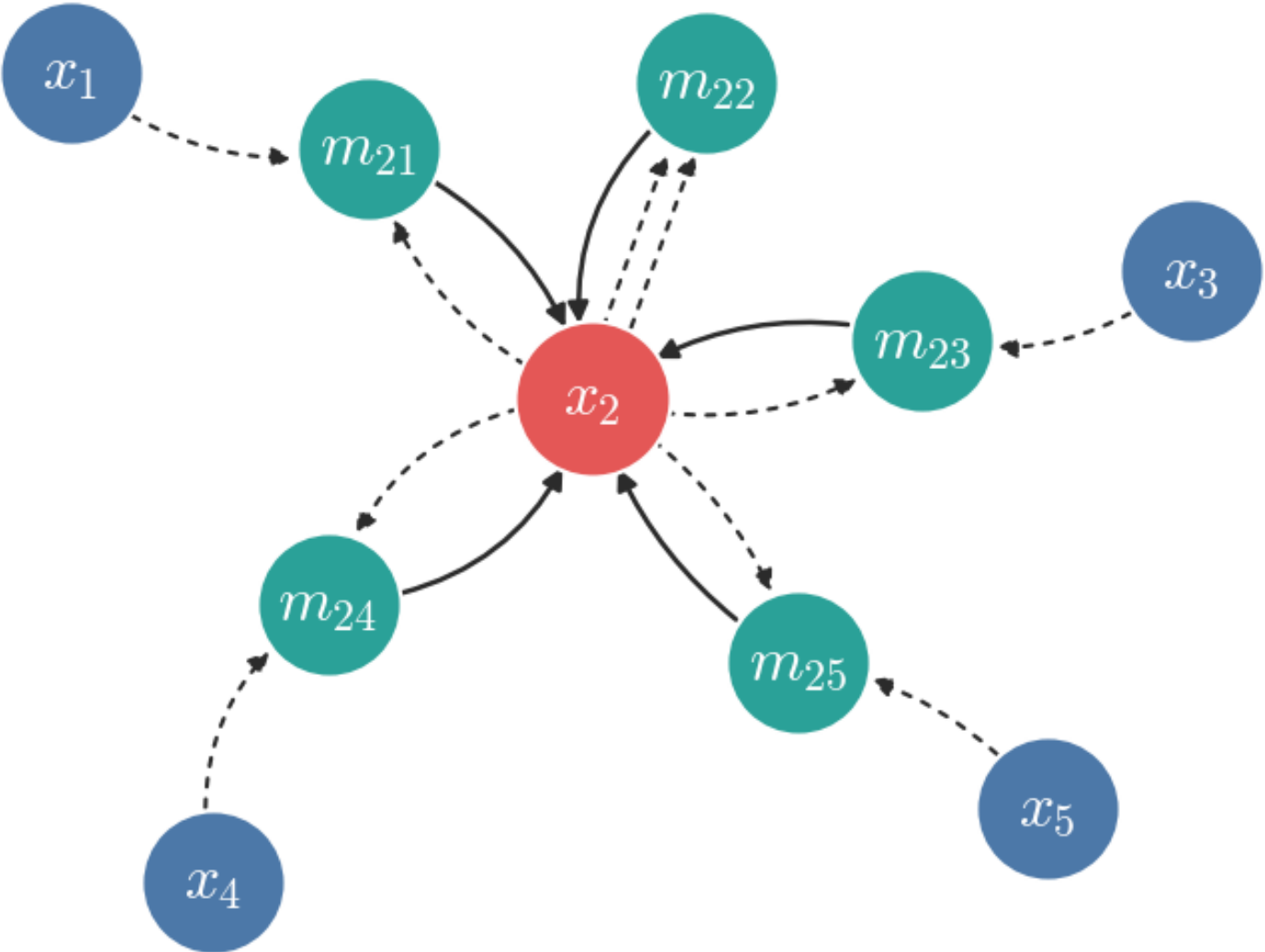
Convolutional

$$h_i = \phi \left(x_i, \bigoplus_{j \in \mathcal{N}_i} c_{ij} \psi(x_j) \right)$$



Attentional

$$h_i = \phi \left(x_i, \bigoplus_{j \in \mathcal{N}_i} \alpha(x_i, x_j) \psi(x_j) \right)$$



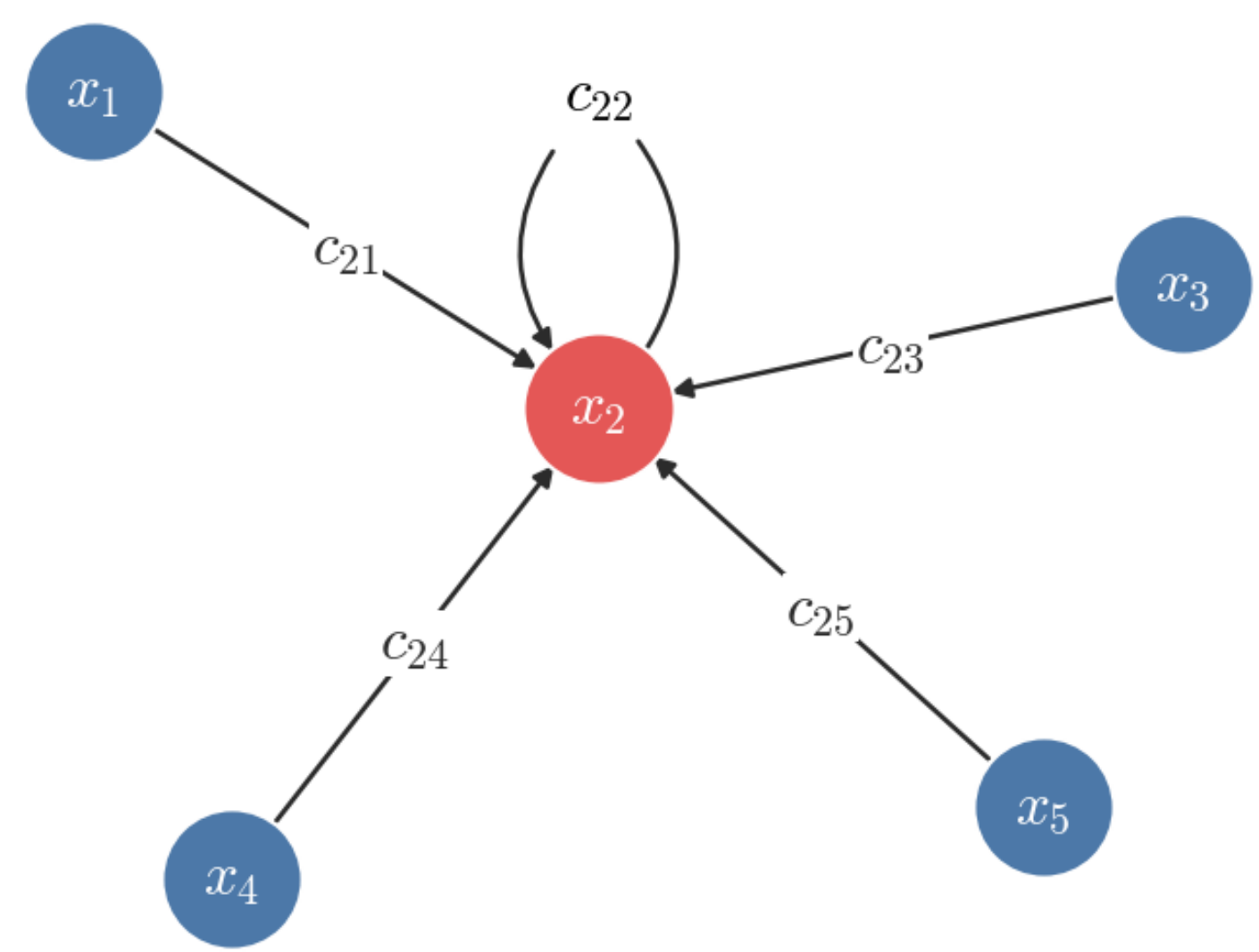
Message-passing

$$h_i = \phi \left(x_i, \bigoplus_{j \in \mathcal{N}_i} \psi(x_i, x_j) \right)$$

Recap: message-passing

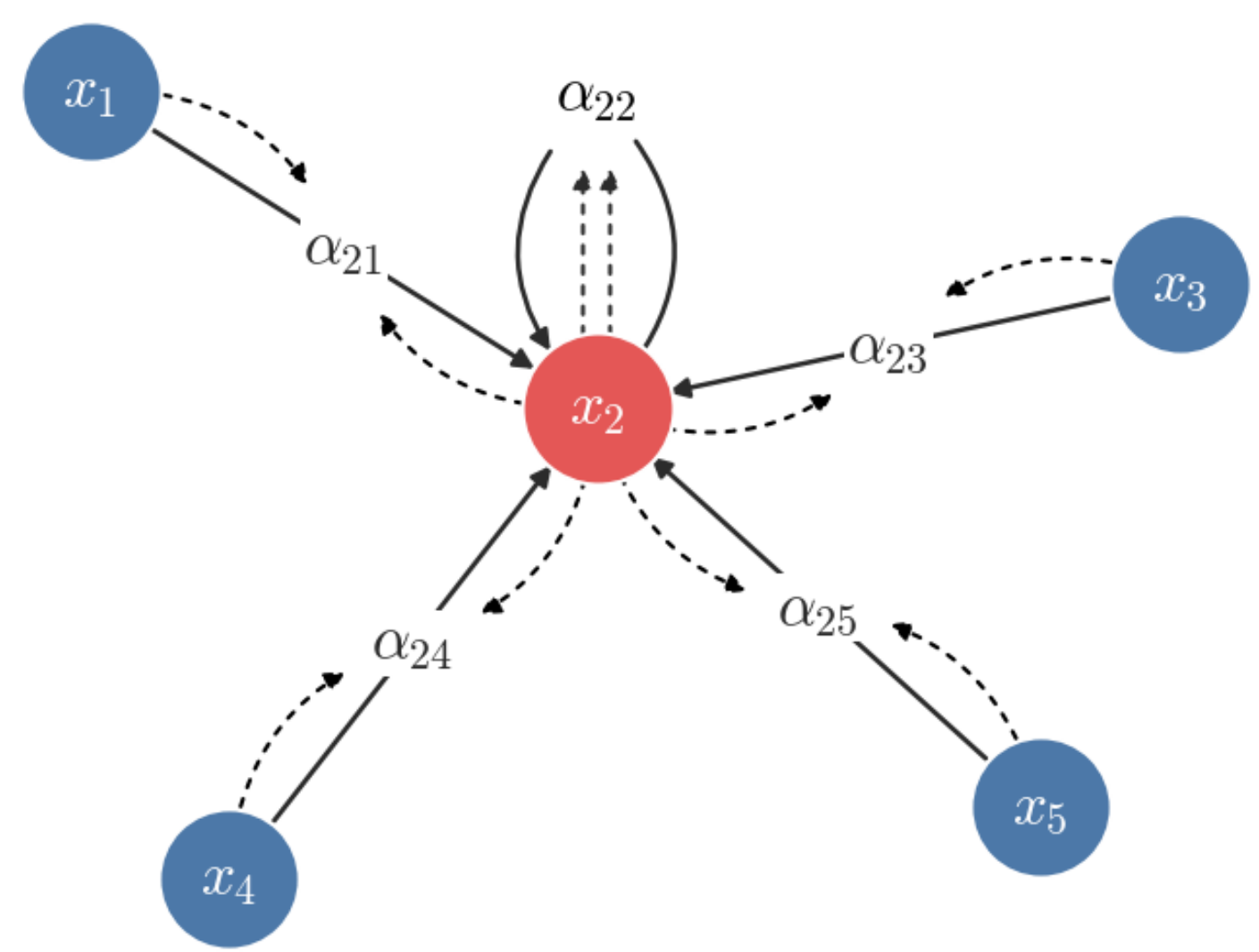
A way of instantiating local functions in GNNs

At each *layer*, we apply this aggregation rule once, updating the features of all nodes



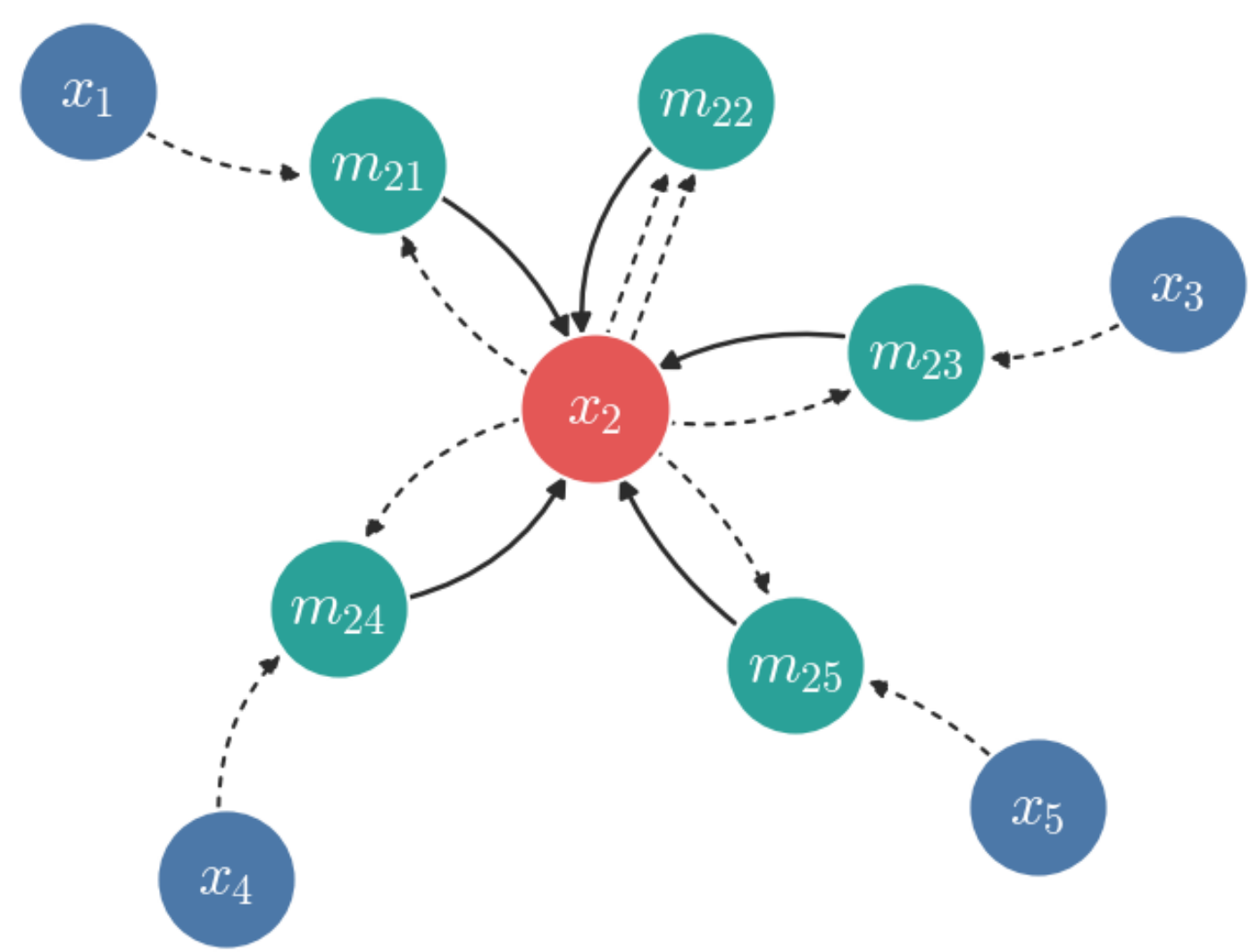
Convolutional

$$h_i^t = \phi_t \left(h_i^{t-1}, \bigoplus_{j \in \mathcal{N}_i} c_{ij}^t \psi_t(h_j^{t-1}) \right)$$



Attentional

$$h_i^t = \phi_t \left(h_i^{t-1}, \bigoplus_{j \in \mathcal{N}_i} \alpha_t(h_i^{t-1}, h_j^{t-1}) \psi_t(h_j^{t-1}) \right)$$

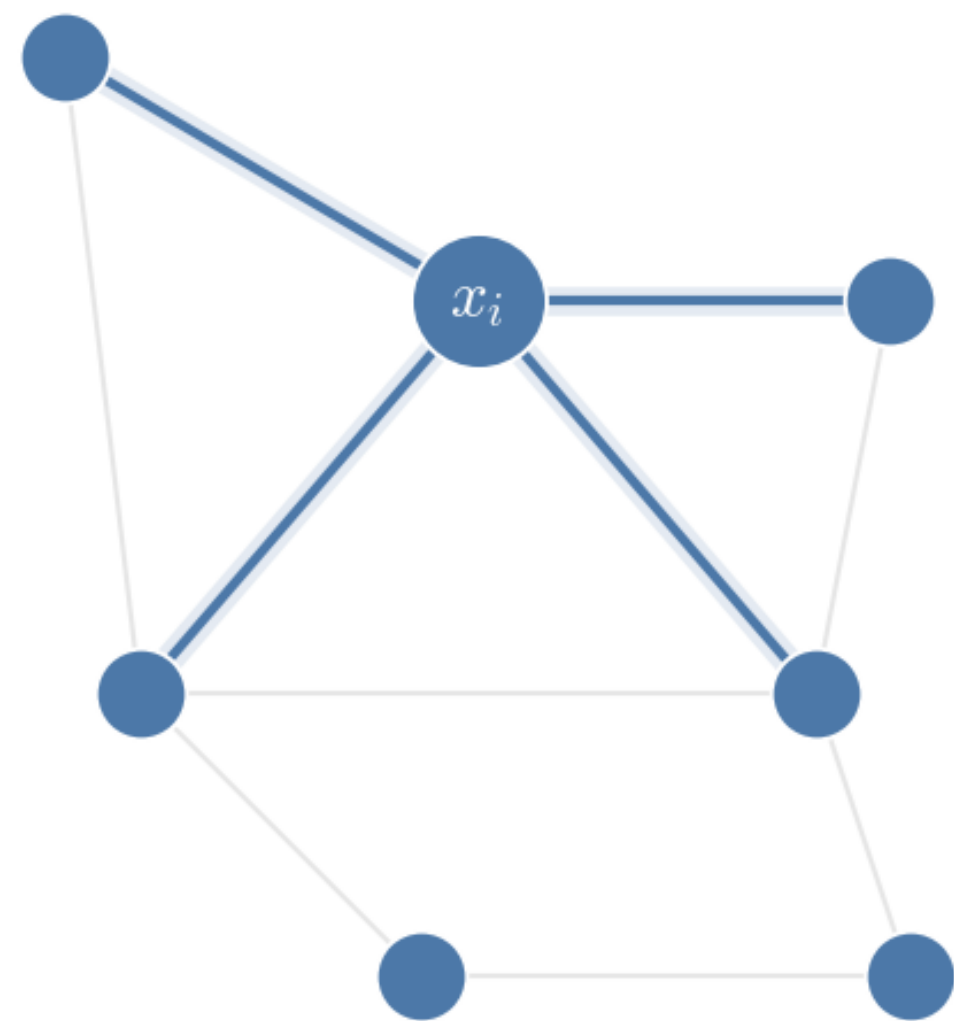


Message-passing

$$h_i^t = \phi_t \left(h_i^{t-1}, \bigoplus_{j \in \mathcal{N}_i} \psi_t(h_i^{t-1}, h_j^{t-1}) \right)$$

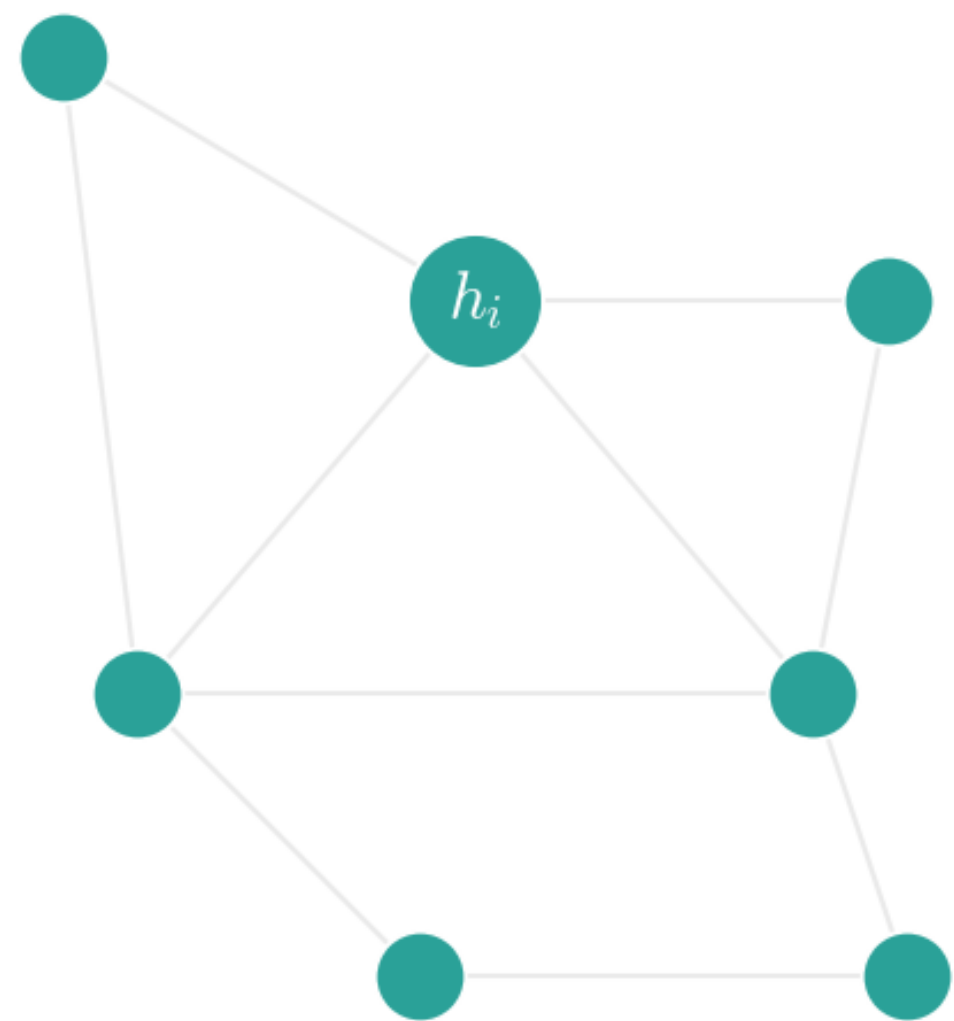
A recipe for GNNs

Solving different tasks



GNN layer

$$f_{\theta}(X, A) = \begin{bmatrix} h_1^t \\ h_2^t \\ \vdots \\ h_n^t \end{bmatrix}$$



Node classification:

$$z_i = f(h_i^L)$$

Graph classification:

$$z_G = f\left(\bigoplus_{i \in V} h_i^L\right)$$

Link prediction:

$$z_{ij} = f(h_i^L, h_j^L, e_{ij})$$

Receptive field in MPNNs

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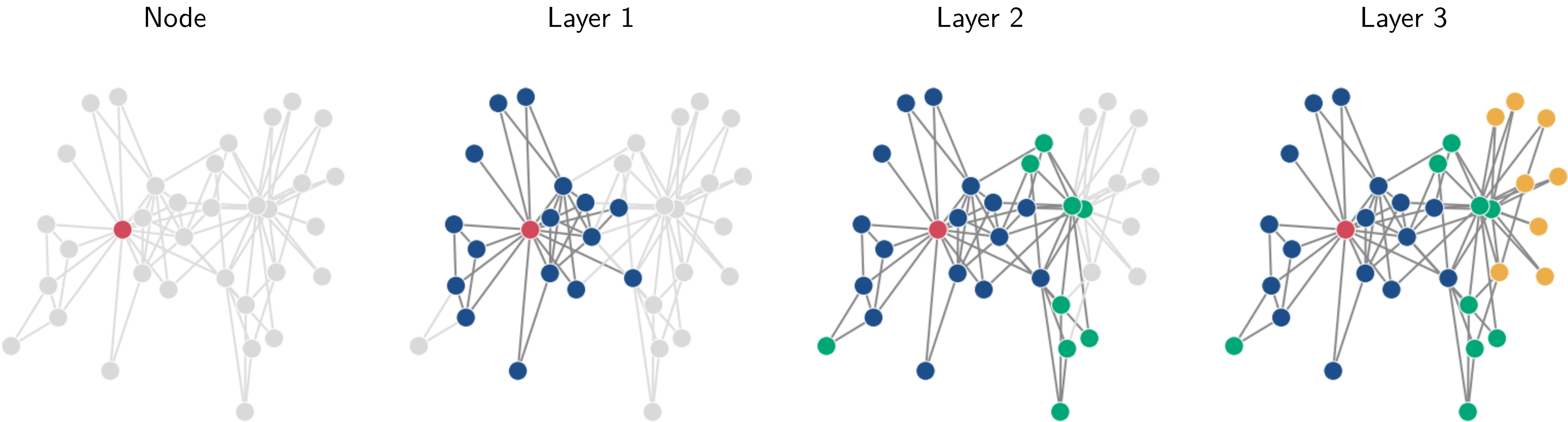
Receptive field in MPNNs

With which nodes can I communicate at each layer?

Def. Receptive field

The receptive field of a node u is the subset of nodes from the graph that have influenced the representation of that node at some layer k

$$\mathcal{N}_u^1 = \mathcal{N}_u \qquad \text{and} \qquad \mathcal{N}_u^k = \mathcal{N}_u^{k-1} \cup \{v : (v, w) \in E \text{ and } w \in \mathcal{N}_u^{k-1}\}$$



Receptive field in MPNNs

Definition in terms of the shortest-path distance

Def. Shortest path distance

Let $G = (V, E)$ be an undirected graph, and $u, v \in V$ two vertices. A path between them is a collection of edges

$$\tau = \{e_i = (u_i, u_{i+1}) : 0 \leq i \leq n - 1\}$$

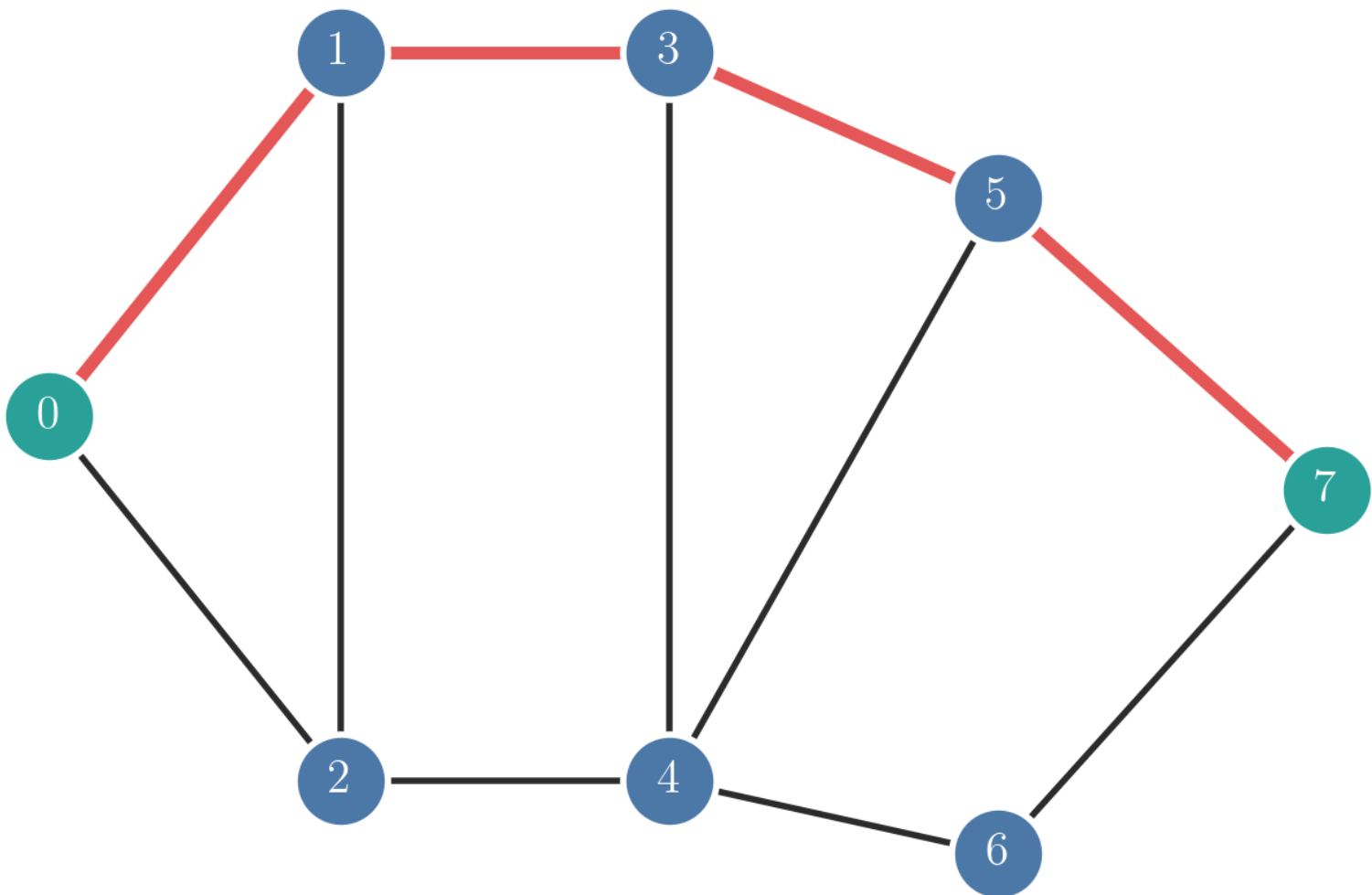
with $u = u_0$ and $v = u_n$

The length of such a path is $n = \text{len}(\tau)$.

The shortest path distance between the vertices is defined as the minimal length among paths between them:

$$d_G : V \times V \rightarrow \mathbb{R}_{\geq 0}$$

$$d_G(u, v) = \min_{\tau \text{ path}} \text{len}(\tau)$$

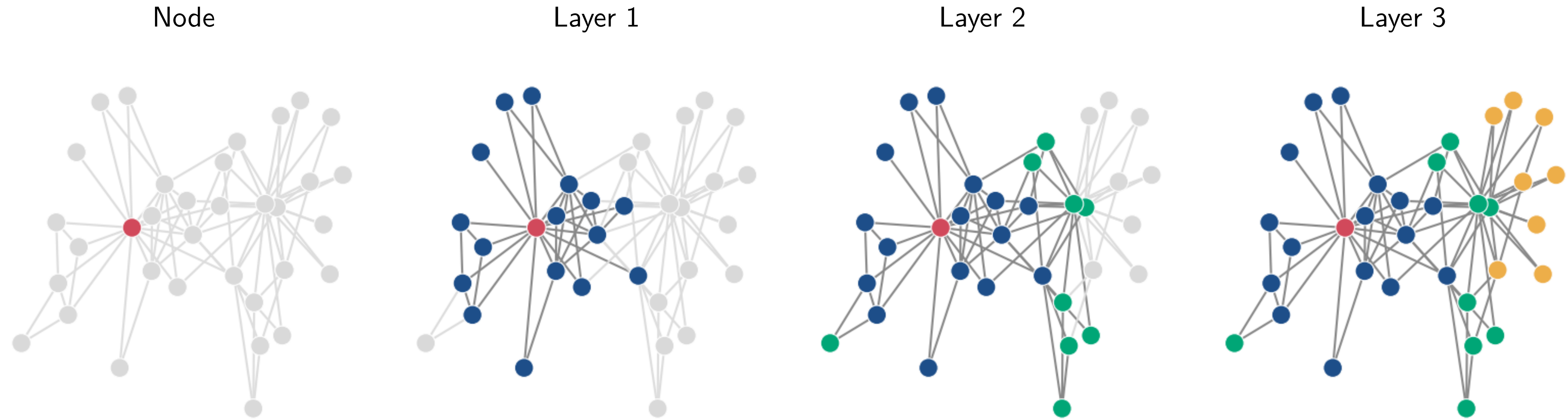


To connect $u = 0$ to $v = 7$ we need:

$$d_G(u, v) = 4$$

Receptive field in MPNNs

Definition in terms of the shortest-path distance



The receptive field of a vertex can also be expressed in terms of the shortest path distance:

$$\mathcal{N}_u^k = \{v \in V : d_G(u, v) \leq k\}$$

This is also called the ***k*-hop ball**, and can also be denoted as $B_k(u)$

To access information from nodes at distance k , we need at least k layers in our MPNN

Receptive field in MPNNs

Computing its complexity

Lemma:

The receptive field has, at worst, exponential growth: $|\mathcal{N}_u^k| = \mathcal{O}(\exp(k))$

Let $\Delta = \max_{v \in V} \{\deg(v)\}$ and $u \in V$ then:

- $|\{v \in V : d(u, v) = 0\}| = 1$
- $|\{v \in V : d(u, v) = 1\}| \leq \Delta$
- $|\{v \in V : d(u, v) = 2\}| \leq \Delta(\Delta - 1)$
- ...
- $|\{v \in V : d(u, v) = j\}| \leq \Delta(\Delta - 1)^{j-1}$

$$\begin{aligned} |\mathcal{N}_u^k| &= \sum_{j=0}^k |\{v \in V : d(u, v) = j\}| \\ &\leq 1 + \sum_{j=1}^k \Delta(\Delta - 1)^{j-1} \\ &\leq 1 + \Delta \frac{(\Delta - 1)^k - 1}{\Delta - 2} \\ &\leq C(\Delta - 1)^k \end{aligned}$$

Using $\Delta \geq 2$ fixed

$$|\mathcal{N}_u^k| = \mathcal{O}((\Delta - 1)^k) = \mathcal{O}(\exp(k \log(\Delta - 1))) = \mathcal{O}(\exp(k))$$

Problem radius

How many layers do I need to solve a task?

Def. Problem radius

The range of interaction r needed to solve a given problem

Example:

- To predict the degree of a node we need $r = 1 \longrightarrow$ short range interaction
- To predict the shortest path distance $d_G(u, v)$ we need as many layers as this distance $\longrightarrow$ long range interaction

For real-world problems, the problem radius is unknown in principle

Community detection tasks (e.g. paper classification on a collaboration network) usually have shorter range

Other tasks such as *predicting molecular properties* like dipoles can depend on longer range interactions

For a given task with radius r we need at least $k \geq r$ layers

How do GNNs perform in these regimes?

Partitioning a graph

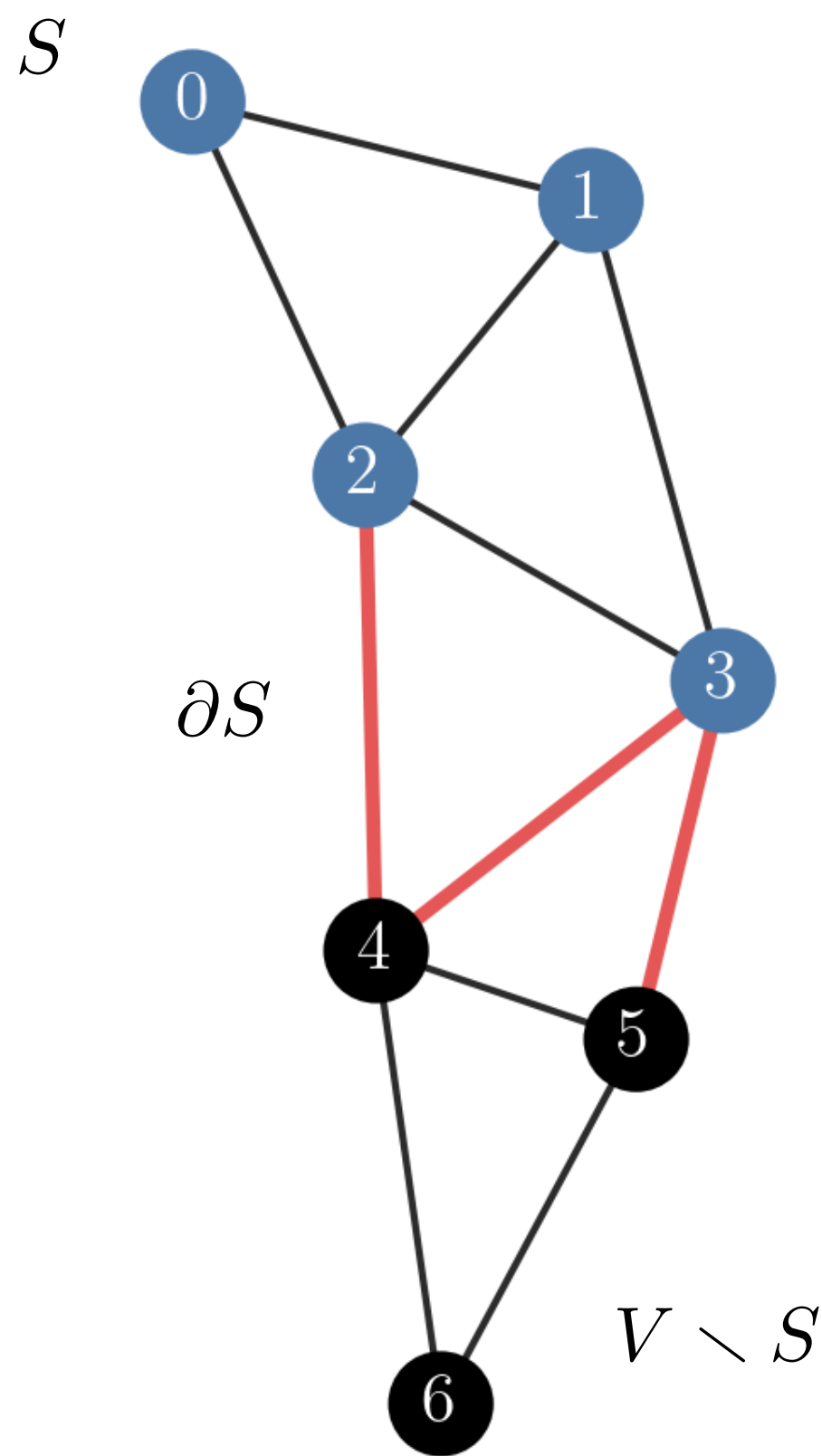
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Isoperimetric ratio

Definition

For $S \subseteq V$ a subset of vertices in a graph $G = (V, E)$, to measure how well separated this subset is from the rest of nodes we can look at the *boundary* of S

$$\partial S = \{(u, v) \in E : u \in S, v \in V \setminus S\}$$



Def. Isoperimetric ratio

For $S \subseteq V$ a subset of vertices in a graph $G = (V, E)$, the isoperimetric ratio is the relative number of its boundary with respect to the size of S

$$\theta(S) = \frac{|\partial S|}{|S|}$$

The **isoperimetric ratio of the graph** is the minimum over the isoperimetric ratio of sets of at least $|V|/2$ vertices

$$\theta_G = \min_{|S| \leq |V|/2} \frac{|\partial S|}{|S|}$$

Isoperimetric ratio and Fiedler's value

Establishing a bound

Recall: for a graph $G = (V, E)$ we defined its (unnormalized) Laplacian as $L = D - A$

For a connected graph, $\lambda_1 = 0$ is an eigenvector and the first non-zero eigenvector $\lambda_2 \neq 0$ is called Fiedler's value

Linear algebra fact:

$$\lambda_2 = \min_{\mathbf{x} : \mathbf{x} \perp \mathbf{1}} \frac{\mathbf{x}^\top L \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}$$

Raighley quotient



Theorem. Bound for the isoperimetric ratio

For every $S \subseteq V$,

$$\theta(S) \geq \lambda_2(1 - s)$$

where $s = |S|/|V|$. In particular,

$$\theta_G \geq \lambda_2/2$$

Isoperimetric ratio and Fiedler's value

Proving the bound

For $\mathbf{x} \neq 0, \mathbf{x} \perp \mathbf{1}$ we have: $\mathbf{x}^\top L \mathbf{x} \geq \lambda_2 \mathbf{x}^\top \mathbf{x}$

To exploit this inequality, we need to find such a vector related to S . Try:

$$\mathbf{x} = \mathbf{1}_S - s\mathbf{1}, \quad \mathbf{1}_S(u) = \begin{cases} 1 & \text{if } u \in S, \\ 0 & \text{otherwise} \end{cases}$$

For which: $\mathbf{x}^\top \mathbf{1} = \mathbf{1}_S^\top \mathbf{1} - s\mathbf{1}^\top \mathbf{1} = |S| - s|V| = 0$

And: $\mathbf{x}^\top L \mathbf{x} = \sum_{(u,v) \in E} ((\mathbf{1}_S(u) - s) - (\mathbf{1}_S(v) - s))^2 = |\partial S|$ $((\mathbf{1}_S(u) - s) - (\mathbf{1}_S(v) - s))^2 = \begin{cases} 1 & u \in S, v \notin S \text{ or } u \notin S, v \in S \\ 0 & \text{otherwise} \end{cases}$

Since $\mathbf{x}^\top \mathbf{x} = |S| - 2s|S| + s^2|V| = |S| - 2\frac{|S|^2}{|V|} + \frac{|S|^2}{|V|} = |S| - \frac{|S|^2}{|V|} = |S|(1 - s)$

We arrive to $\frac{|\partial S|}{|S|} = \theta(S) \geq \lambda_2(1 - s)$

And since $|S| \leq \frac{n}{2} \iff s \leq \frac{1}{2} \iff 1 - s \geq \frac{1}{2}$ we conclude $\theta_G \geq \frac{\lambda_2}{2}$

Conductance

Definition

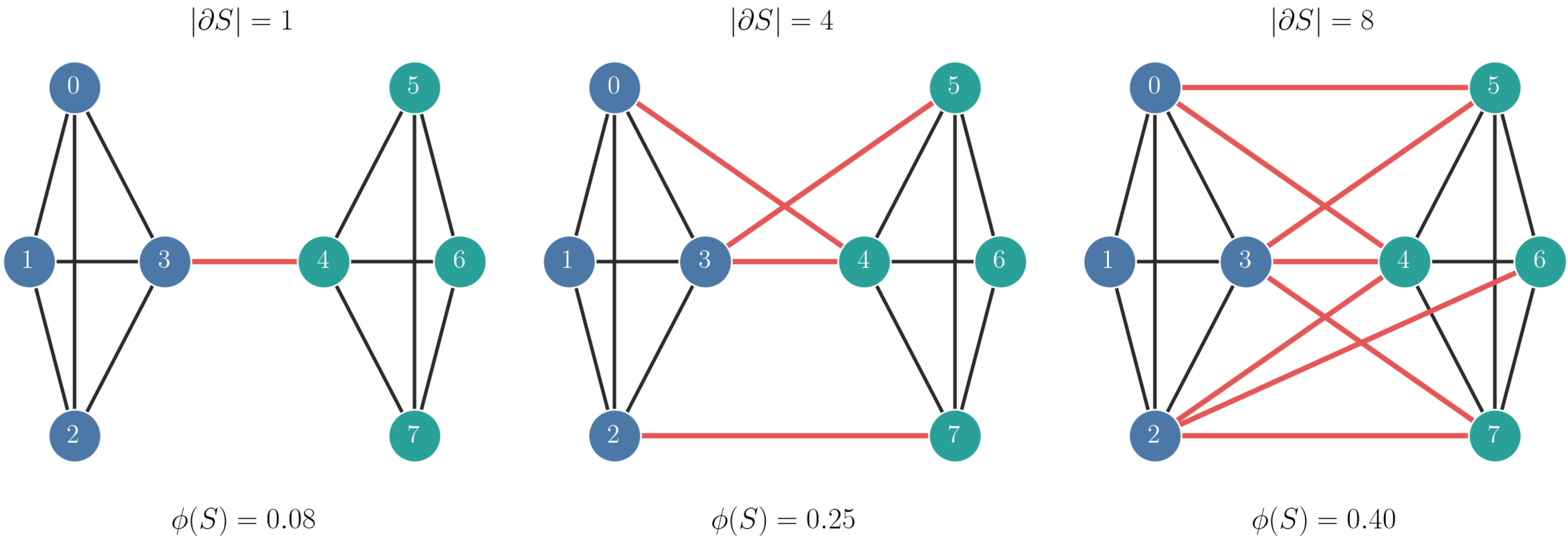
Def. Conductance

For $S \subseteq V$ a subset of vertices in a graph $G = (V, E)$, the conductance of S is

$$\phi(S) = \frac{|\partial S|}{\min\{d(S), d(V \setminus S)\}}, \text{ where } d(S) = \sum_{u \in S} d_u$$

The conductance of the graph is

$$\phi_G = \min_{S \subseteq V} \phi(S)$$



Normalized Laplacian

Motivation and definition

Def. Conductance

$$\phi(S) = \frac{|\partial S|}{\min\{d(S), d(V \setminus S)\}}, \text{ where } d(S) = \sum_{u \in S} d_u$$

For the isoperimetric ratio bound, we found the Rayleigh quotient very useful. In this case, we observe:

$$\frac{\mathbf{1}_S^\top L \mathbf{1}_S}{\mathbf{1}_S^\top D \mathbf{1}_S} = \frac{\sum_{(u,v) \in E} (\mathbf{1}_S(u) - \mathbf{1}_S(v))^2}{\sum_{u \in S} d_u} = \frac{|\partial S|}{d(S)}$$

So we might be interested in the following generalized Rayleigh quotient:

$$\frac{\mathbf{y}^\top L \mathbf{y}}{\mathbf{y}^\top D \mathbf{y}} \xrightarrow{\mathbf{y} = D^{-1/2} \mathbf{x}} \frac{\mathbf{x}^\top D^{-1/2} L D^{-1/2} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}$$

Def. Normalized Laplacian

$$N = D^{-1/2} L D^{-1/2}$$

Normalized Laplacian

Eigenvectors and conductance

Def. Normalized Laplacian

$$N = D^{-1/2} L D^{-1/2}$$

It is also symmetric and positive semidefinite, so it has $n = |V|$ positive eigenvalues:

$$0 = \nu_1 \leq \nu_2 \leq \cdots \leq \nu_n \leq 2$$

0 is an eigenvalue with eigenvector $\mathbf{d} = (\sqrt{d_1}, \dots, \sqrt{d_n})$ since $D^{-1/2} \mathbf{d} = \mathbf{1}$ so $N \mathbf{d} = 0$

Linear algebra fact:

$$\nu_2 = \min_{\mathbf{y} : \mathbf{y} \perp \mathbf{d}} \frac{\mathbf{y}^\top L \mathbf{y}}{\mathbf{y}^\top D \mathbf{y}}$$

Theorem. Bound for the conductance

For every $S \subseteq V$,

$$\nu_2 \leq d(V) \frac{|\partial S|}{d(S)d(V \setminus S)}$$

So

$$\phi(S) \geq \frac{\nu_2}{2}$$

The spectral method and Cheeger inequalities

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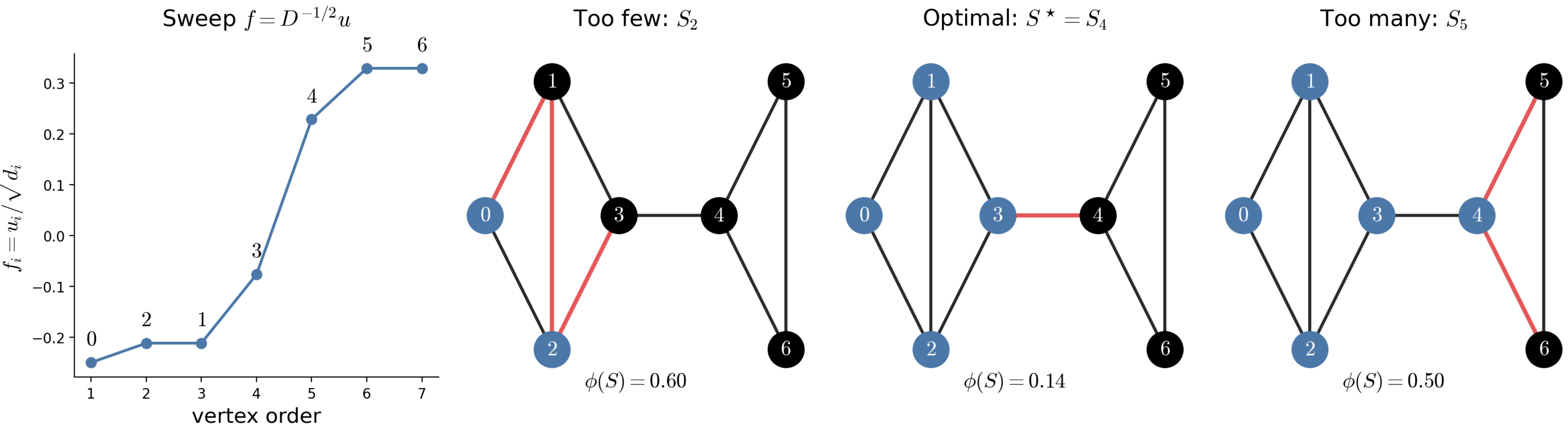
The spectral method

Definition

Let $G = (V, E)$ be a graph with $n = |V|$ vertices

- Let $\mathbf{u} \in \mathbb{R}^n$ be the eigenvector associated to ν_2 , the first non-zero eigenvalue of N
- Let $\mathbf{f} = D^{-1/2}\mathbf{u}$, so that $L\mathbf{f} = \nu_2\mathbf{Df}$
- Sort the vertices i of G in increasing order of f_i : $f_1 \leq f_2 \leq \dots \leq f_n$
- For each vertex, consider $S_i = \{1, 2, \dots, i\}$ and compute its conductance
- Choose the one with minimum conductance

Example:



Cheeger's Inequalities

Definition

How good is this partition?

Def. Cheeger's Inequality

Let G be a connected graph with conductance ϕ_G . Let $\gamma = \phi(S^\star)$ denote the conductance of the optimal partition found from the spectral algorithm applied to the *normalized Laplacian*. Then

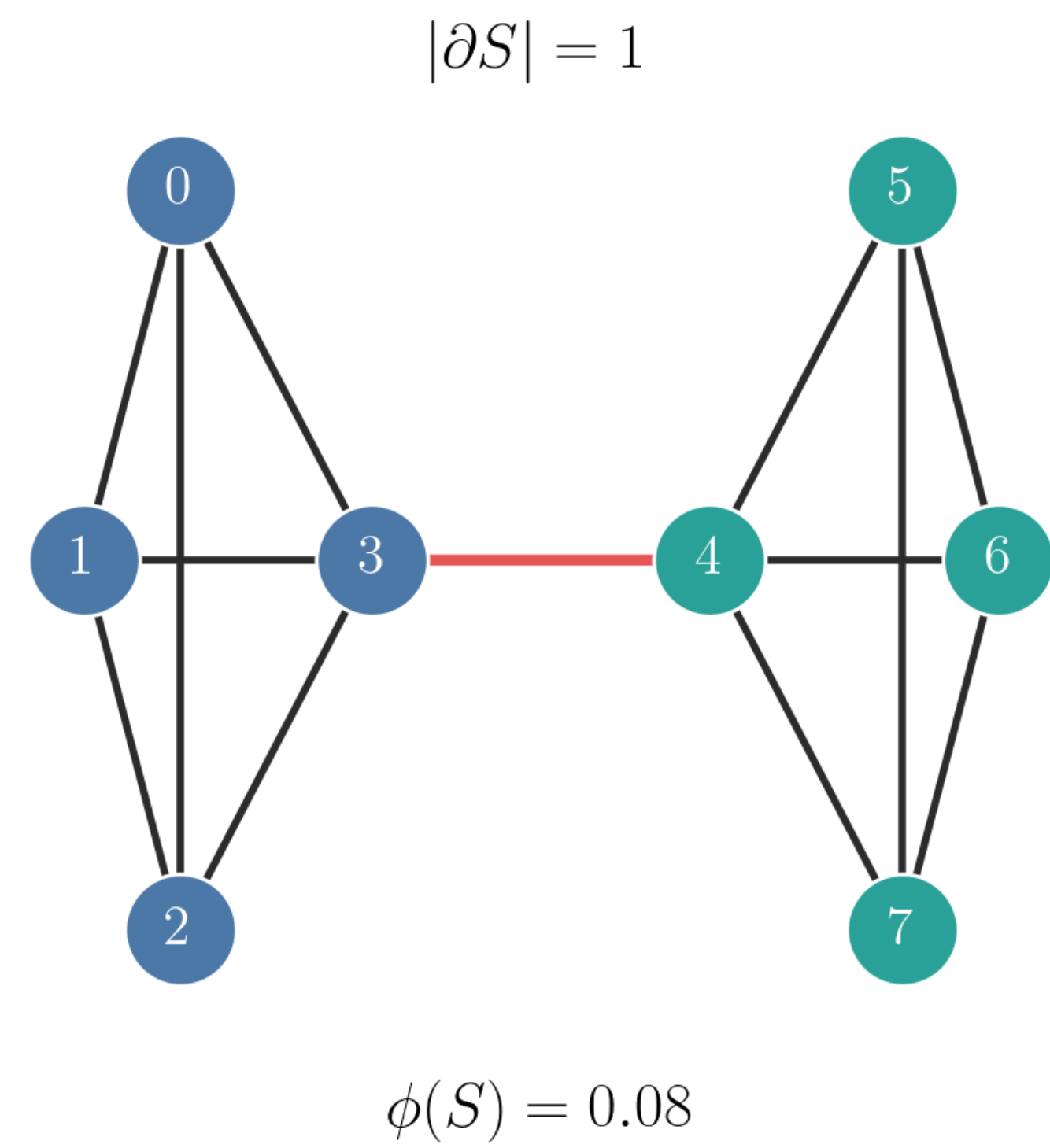
$$\frac{\nu_2}{2} \leq \phi_G \leq \gamma \leq \sqrt{2\nu_2}$$

Fiedler's value (for the normalized Laplacian) gives us a bound on “how disconnected” our graph can be (the minimal conductance that the graph can achieve)

Recall the interpretation that we saw of Fiedler's value on the second lecture, this explains that initial intuition

Bottlenecks

Definition



When we have a graph with two *communities* (subsets of nodes tightly interconnected) connected by few edges, we say that we have a ***bottleneck***

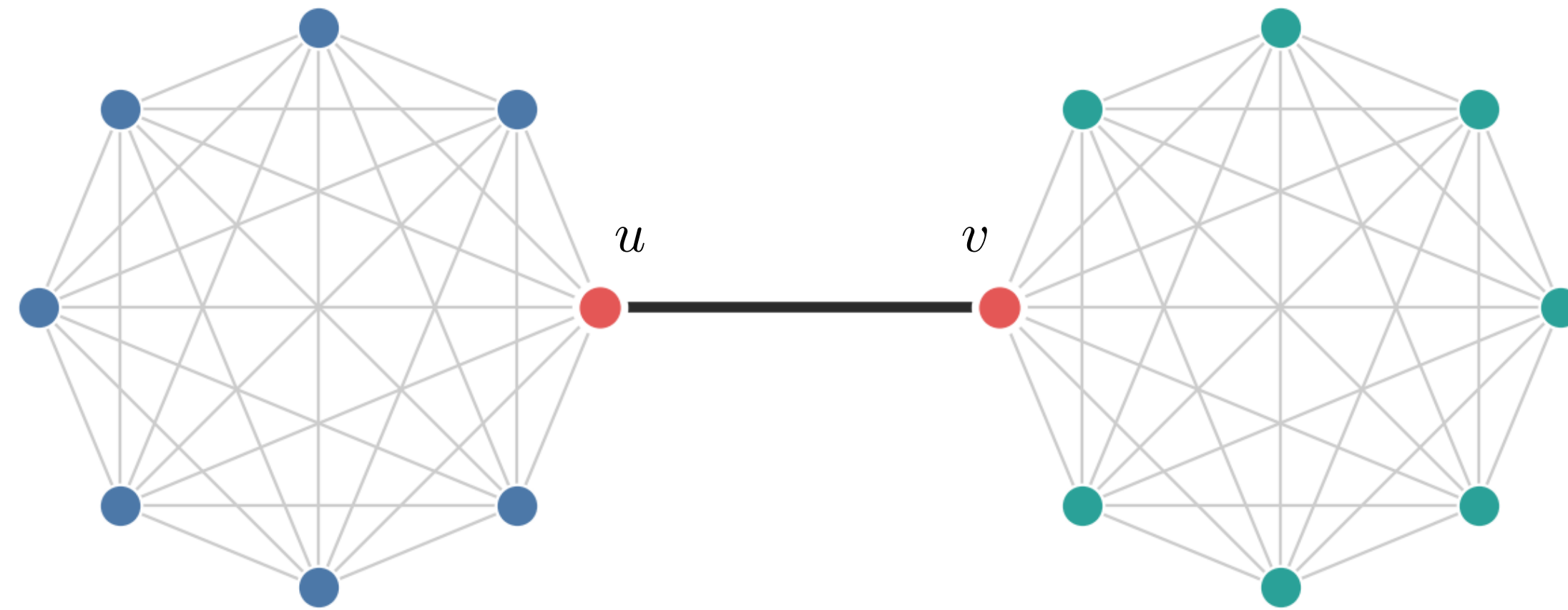
As we have seen, this can be characterized by low graph conductance

Over-squashing

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Over-squashing

The problem of bottlenecks in message-passing



Message-passing:

$$h_i^t = \phi_t \left(h_i^{t-1}, \bigoplus_{j \in \mathcal{N}_i} \psi_t(h_i^{t-1}, h_j^{t-1}) \right)$$

- At layer t , each node receives messages from at most

$$|\mathcal{N}_u^t| = \mathcal{O}(\exp(t))$$

- When we have a bottleneck between two communities, to communicate between them, all the information is compressed in a single message of fixed size across the edge of the bottleneck

$$\psi_t(h_u^{t-1}, h_v^{t-1}) \in \mathbb{R}^m$$

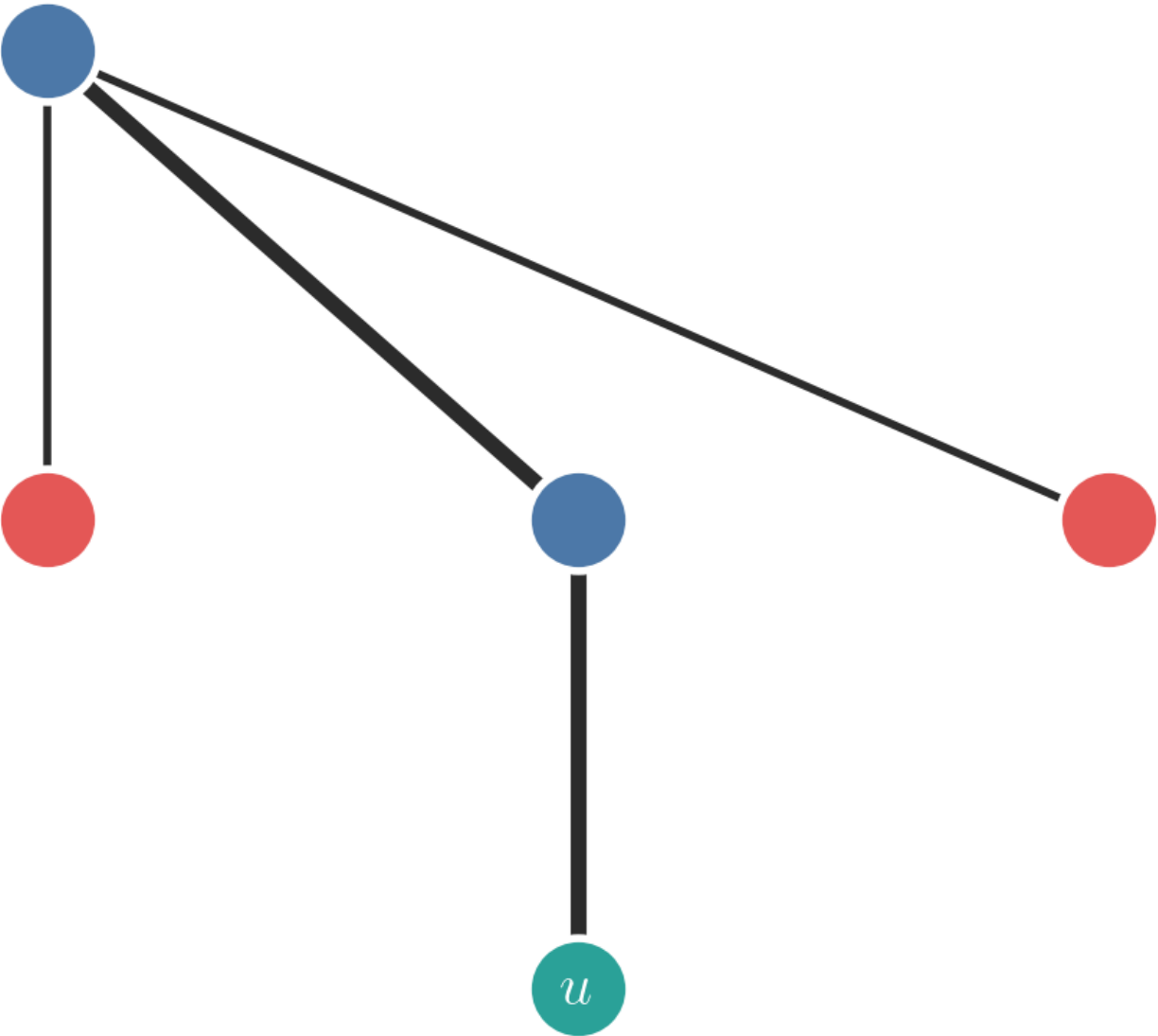
As a result, the messages from the nodes in one community are condensed into a single message, potentially losing information

Alon, Uri, and Eran Yahav. "On the Bottleneck of Graph Neural Networks and Its Practical Implications." *International Conference on Learning Representations* (2020).

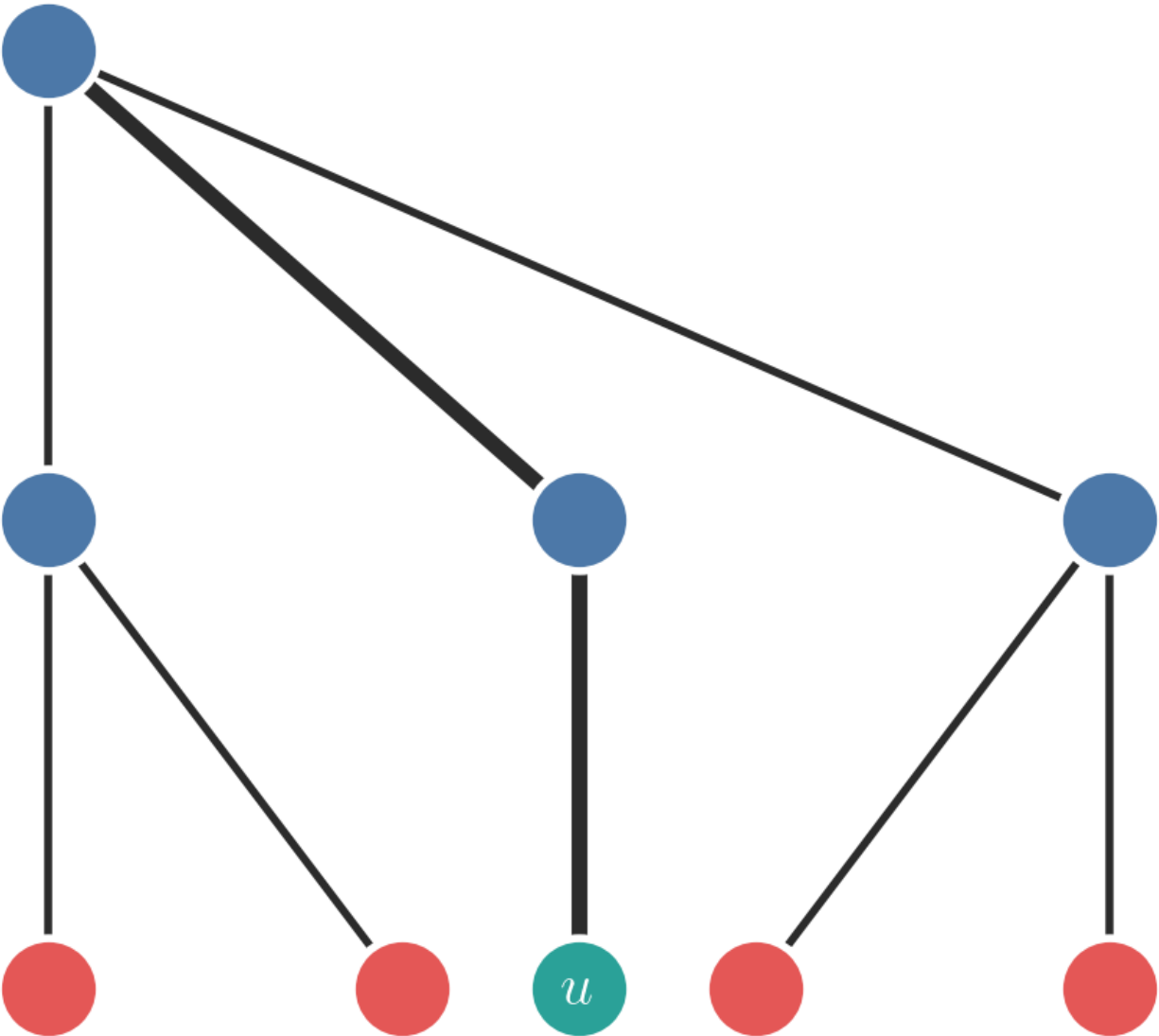
Over-squashing

Example

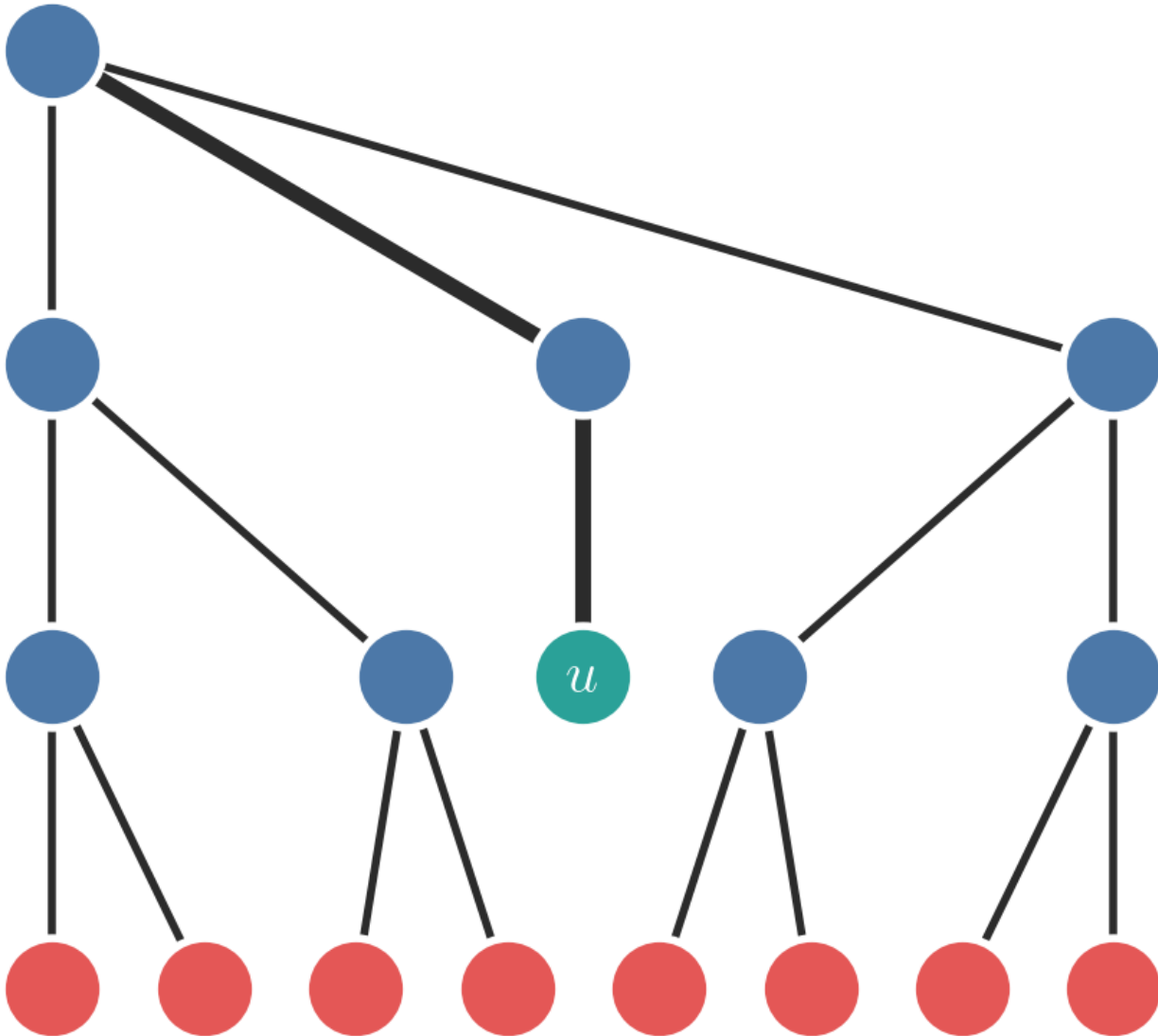
NeighborsMatch problem: imagine you want to predict a property for u that depends on the nodes in red, for instance, a label depending on the number of neighbors



$$|\mathcal{S}| = 2^1 = 2 \quad d_G(\mathcal{S}, u) = 3$$



$$|\mathcal{S}| = 2^2 = 4 \quad d_G(\mathcal{S}, u) = 4$$



$$|\mathcal{S}| = 2^3 = 8 \quad d_G(\mathcal{S}, u) = 5$$

Over-squashing

Example

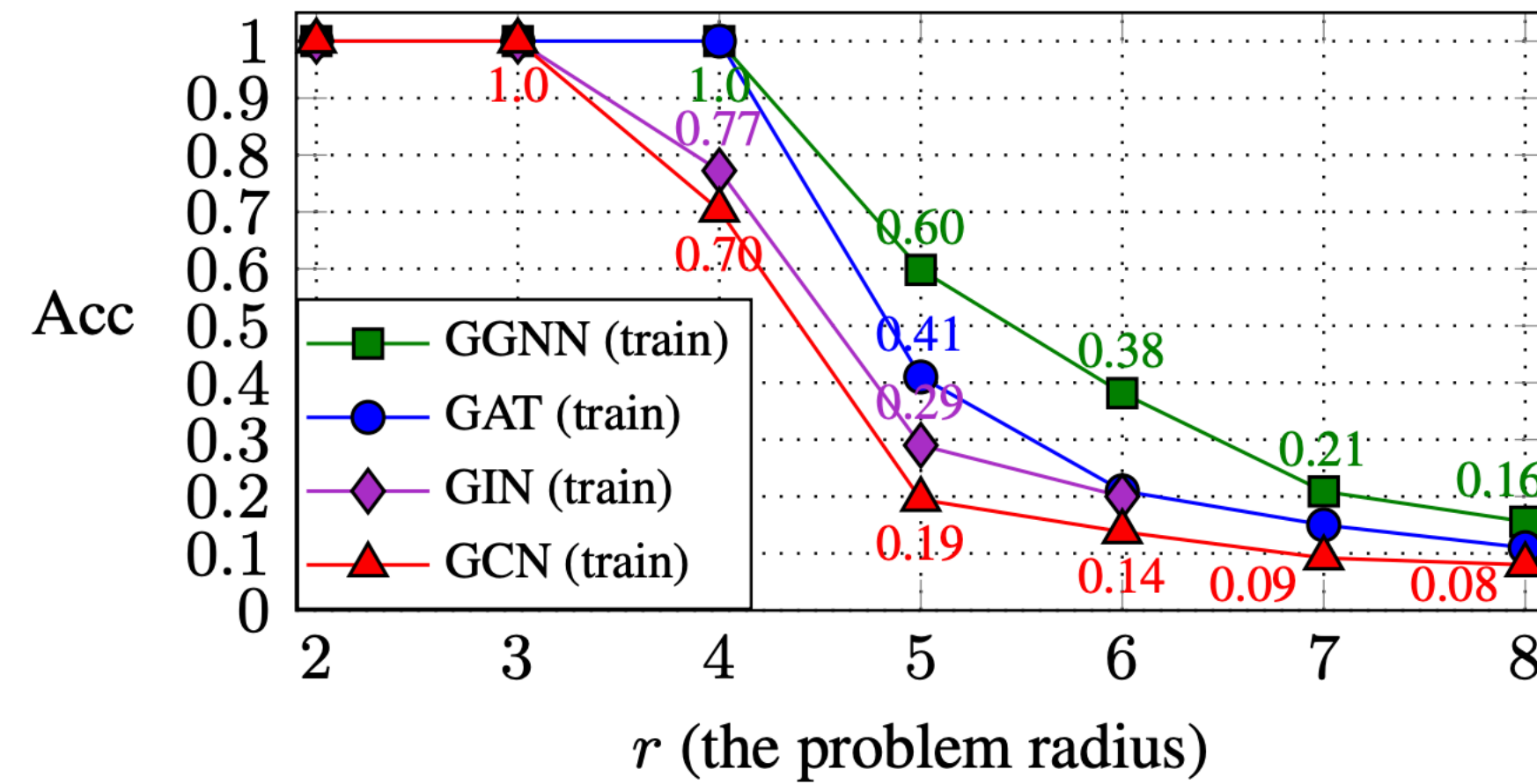


Figure 3: Accuracy across *problem radius* (tree depth) in the NEIGHBORMATCH problem. Over-squashing starts to affect GCN and GIN even at $r = 4$.

As a result of the bottleneck connecting the rest of the tree with u , GNNs fail to propagate messages originating from the source set, and perform poorly

This happens in general for tasks depending on long range interactions in graphs with bottlenecks

Over-squashing

The Jacobian as a proxy for oversquashing

The features of a node at a certain layer depend on the initial features $h_i^{(t)} = h_i^{(t)}(\mathbf{x}_1, \dots, \mathbf{x}_n) \in \mathbb{R}^m$, as a differentiable function as long as the update ϕ_t and the message-passing ψ_t functions are differentiable

$$h_i^{(t)} : \mathbb{R}^{n \times k} \rightarrow \mathbb{R}^m, \quad \begin{pmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix} \mapsto h_i^{(t)} = ((h_i^{(t)})^1, \dots, (h_i^{(t)})^m)$$

Over-squashing can be understood as the representation of i failing to be affected by the features of node j at distance r , and this is captured by the **Jacobian matrix**:

$$\frac{\partial h_i^{(r)}}{\partial \mathbf{x}_j} = \begin{pmatrix} \frac{\partial (h_i^{(r)})^1}{\partial \mathbf{x}_j^1} & \frac{\partial (h_i^{(r)})^1}{\partial \mathbf{x}_j^2} & \cdots & \frac{\partial (h_i^{(r)})^1}{\partial \mathbf{x}_j^k} \\ \frac{\partial (h_i^{(r)})^2}{\partial \mathbf{x}_j^1} & \frac{\partial (h_i^{(r)})^2}{\partial \mathbf{x}_j^2} & \cdots & \frac{\partial (h_i^{(r)})^2}{\partial \mathbf{x}_j^k} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial (h_i^{(r)})^m}{\partial \mathbf{x}_j^1} & \frac{\partial (h_i^{(r)})^m}{\partial \mathbf{x}_j^2} & \cdots & \frac{\partial (h_i^{(r)})^m}{\partial \mathbf{x}_j^k} \end{pmatrix} \in \mathbb{R}^{m \times k}$$

$$h_i^{(r)} = ((h_i^{(r)})^1, \dots, (h_i^{(r)})^m) \in \mathbb{R}^m$$

$$\mathbf{x}_j = (\mathbf{x}_j^1, \dots, \mathbf{x}_j^k) \in \mathbb{R}^k$$

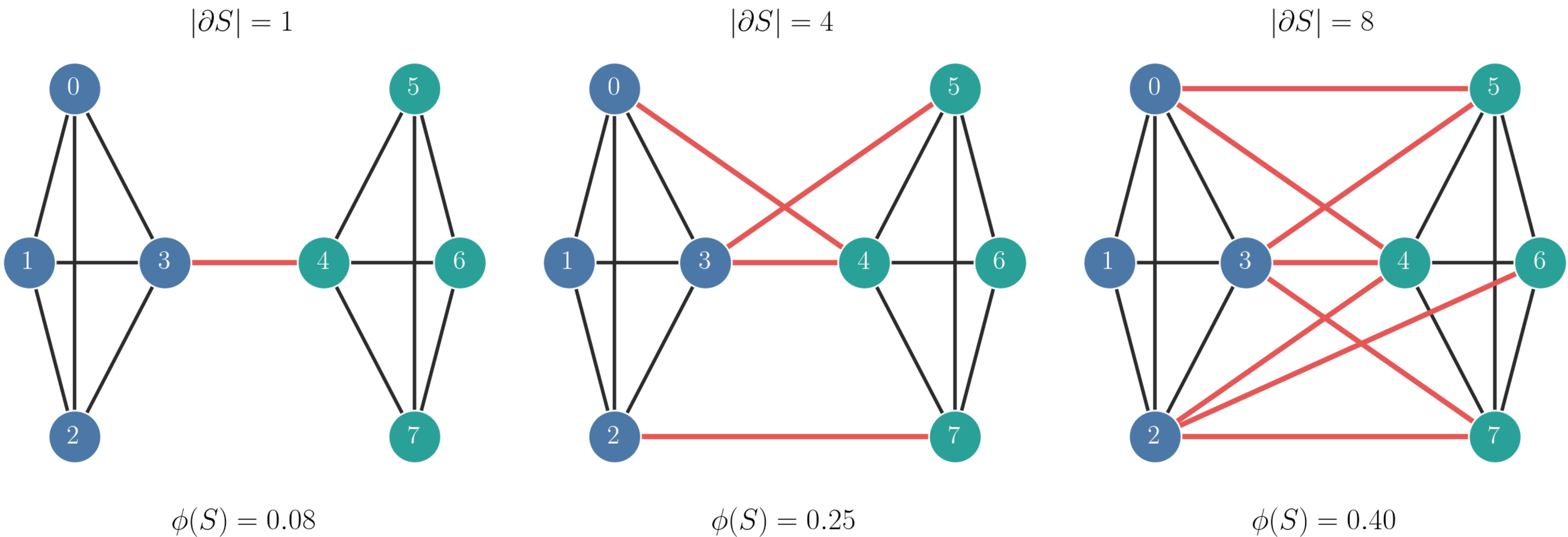
Topping, Jake, Francesco Di Giovanni, Benjamin Paul Chamberlain, Xiaowen Dong, and Michael M. Bronstein. Understanding Over-Squashing and Bottlenecks on Graphs via Curvature. *International Conference on Learning Representations* (2021).

Rewiring

A mitigation for over-squashing

Try to identify the presence of bottlenecks between communities in your graphs

Add edges between the communities that are strained to increase the communication between them



Topping, J., Giovanni, F. D., Chamberlain, B. P., Dong, X., & Bronstein, M. M. (2021). *Understanding over-squashing and bottlenecks on graphs via curvature*. International Conference on Learning Representations.

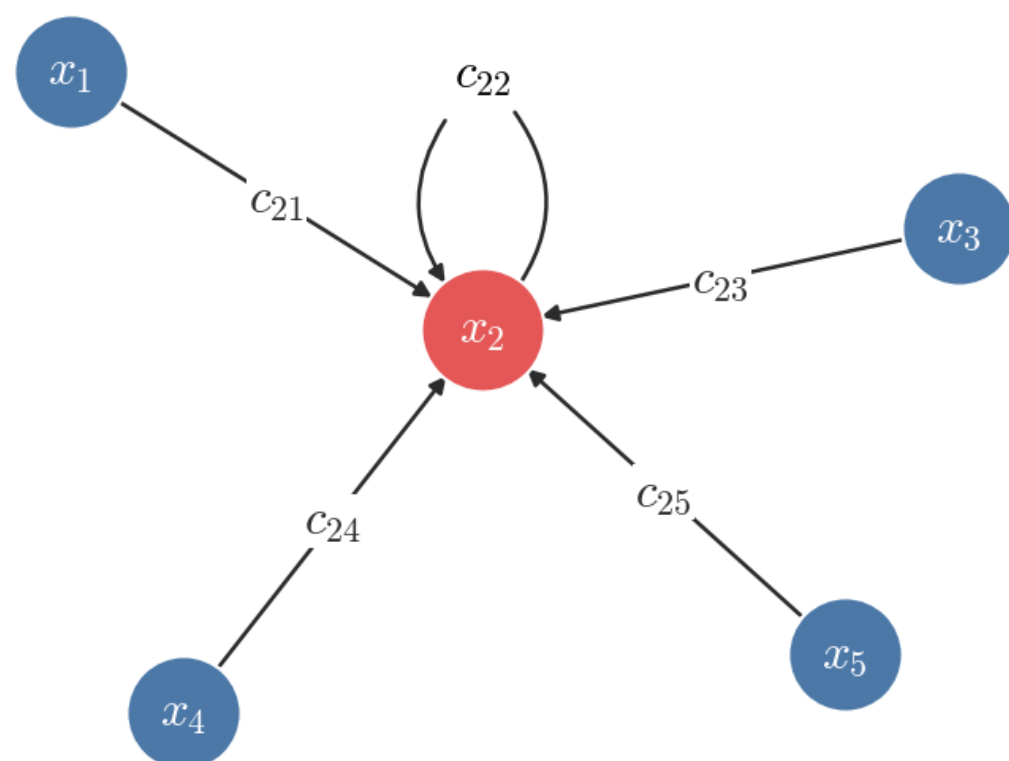
Black, M., Wan, Z., Nayyeri, A., & Wang, Y. (2023). Understanding Oversquashing in GNNs through the Lens of Effective Resistance. *Proceedings of the 40th International Conference on Machine Learning*, 2528–2547.

Over-smoothing

Session 3 — Learning on Topological Data

Over-smoothing

GCNNs revisited



Update rule of GCN¹:

$$h_i^{t+1} = \phi \left(h_i^t, \bigoplus_{j \in \mathcal{N}_i} c_{ij} \psi(h_j^t) \right) = \sum_{j \in \mathcal{N}_i \cup \{i\}} (d_i + 1)^{-1/2} (d_j + 1)^{-1/2} W^{(t)} h_j^t$$

$$\phi = \text{Id} \qquad c_{ij} = (d_i + 1)^{-1/2} (d_j + 1)^{-1/2} \qquad \bigoplus = \sum \qquad \psi(h_j^t) = W^{(t)} h_j^t$$

In matrix form:

$$H^{t+1} = (D + I)^{-1/2} (A + I) (D + I)^{-1/2} H^t W^{(t)} = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^t W^{(t)}$$

Augmented adjacency and degree matrices: adding self loops in all the nodes, so that they can send messages to themselves

$$\begin{aligned} \tilde{A} &= A + I \\ \tilde{D} &= D + I \end{aligned} \implies \tilde{L} = \tilde{D} - \tilde{A} = (D + I) - (A + I) = D - A = L$$

This leaves the Laplacian unchanged

However the normalized Laplacian does change!

$$\tilde{N} = \tilde{D}^{-1/2} L \tilde{D}^{-1/2} = I - \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$$

$$H^{t+1} = (I - \tilde{N}) H^t W^{(t)}$$

¹Kipf, Thomas N., and Max Welling. “Semi-Supervised Classification with Graph Convolutional Networks.” *International Conference on Learning Representations* (2017).

Over-smoothing

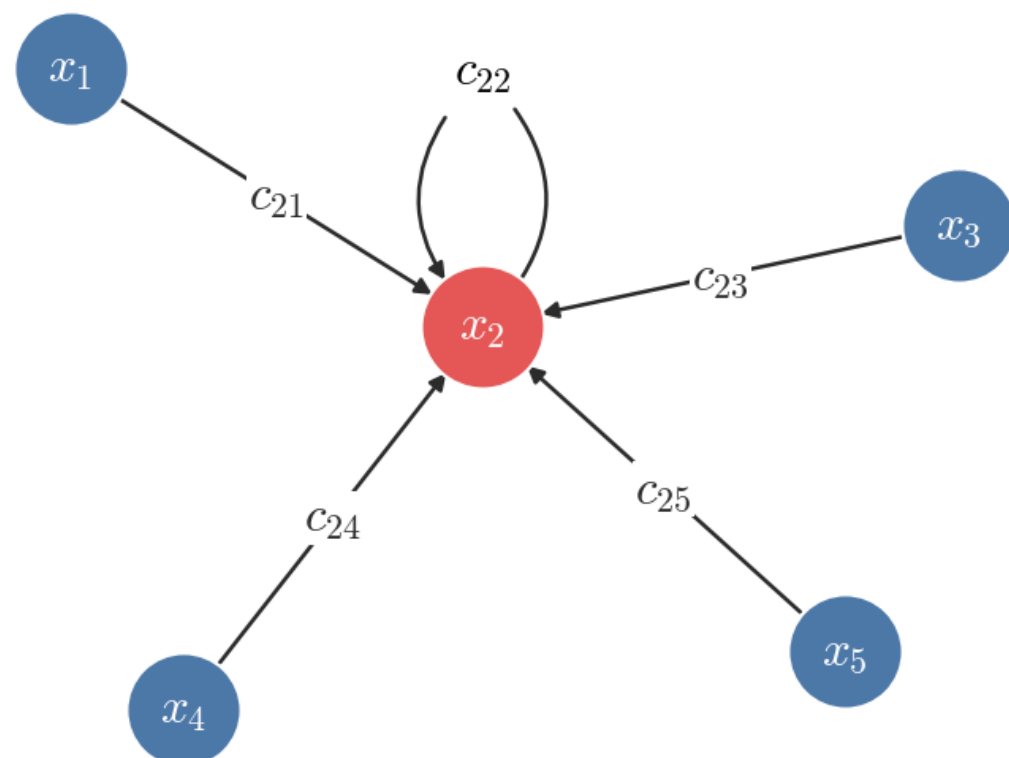
GCNNs revisited

Recall from **Lecture 2** that a signal over the vertices could be expressed as

$$\mathbf{f} = (f_1, \dots, f_n) \in \mathbb{R}^n \quad f_i = f(i) \text{ for } i \in V$$

And that the Laplacian could be seen as an operator over this space

We can see scalar features as signals, so that a GNN is trying to learn a signal over the vertices



$$\mathbf{x}_i = x_i \in \mathbb{R}, \quad X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^{n \times 1}$$

Express the node signal over in the eigenbasis of the normalized Laplacian:

$\tilde{N} = \tilde{D}^{-1/2} L \tilde{D}^{-1/2} = I - \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$ is symmetric and positive semi-definite, so we can find a basis of orthonormal eigenvectors

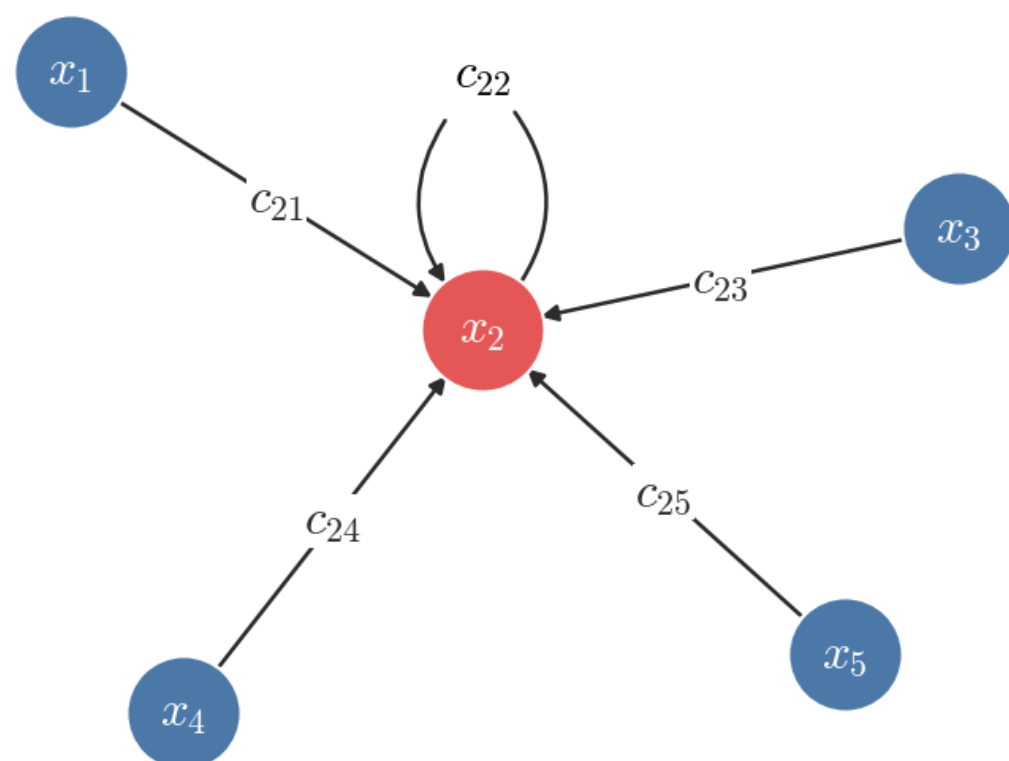
$$\{\Phi_i : i \in V\} \quad \tilde{N} \Phi_i = \tilde{\nu}_i \Phi_i \quad 0 = \tilde{\nu}_1 \leq \tilde{\nu}_2 \leq \dots \leq \tilde{\nu}_n < 2$$

We can write the input node signal in this basis (recall the Graph Fourier Transform)

$$X = \sum_{i=1}^n a_i \Phi_i \in \mathbb{R}^n$$

Over-smoothing

GCNNs revisited



Apply the update rule:

$$H^{t+1} = (I - \tilde{N})H^t \quad W^{(t)} = \text{Id}$$

First layer: $H^1 = (I - \tilde{N})X = \sum_{i=1}^n a_i (I - \tilde{N})\Phi_i = \sum_{i=1}^n a_i (1 - \tilde{\nu}_i)\Phi_i$

Layer t : $H^t = (I - \tilde{N})H^{t-1} = (I - \tilde{N})^t X = \sum_{i=1}^n a_i (1 - \tilde{\nu}_i)^t \Phi_i$

Since the eigenvalues are all positive and strictly less than 2, we have:

$$\tilde{\nu}_1 = 0, \quad 0 < \tilde{\nu}_i < 2, \quad 1 \leq i \leq n \quad \Longleftrightarrow \quad 1 - \tilde{\nu}_1 = 1, \quad -1 < 1 - \tilde{\nu}_i < 1, \quad 1 \leq i \leq n$$

And, as we apply layers in our model:

$$(1 - \tilde{\nu}_i)^t \xrightarrow{t \rightarrow \infty} 0 \quad H^t \xrightarrow{t \rightarrow \infty} a_1 \Phi_1 = \Phi_1 \Phi_1^\top X, \quad \Phi_1 = \frac{\tilde{D}^{1/2} \mathbf{1}}{\sqrt{\sum_i (d_i + 1)}}$$

In other words, the learned representations converge to a **constant shared value** over all the nodes in the graph, which is completely uninformative and will not be useful for learning tasks!

Over-smoothing

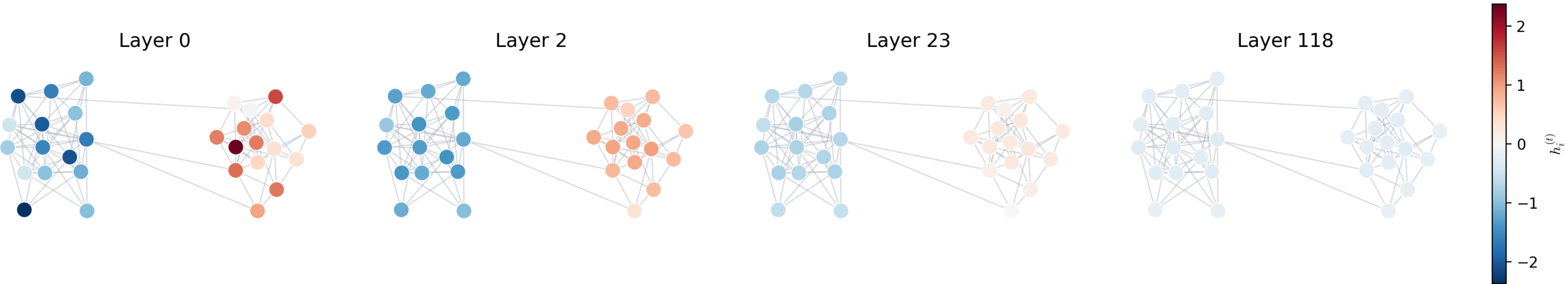
Representation collapse

This phenomenon happens in general in GNNs, and it is known as over-smoothing

However, the smoothing can actually be useful before it becomes harmful

Example: $x_i = y_1 + \epsilon_i, \quad y_i \in \{1, -1\}, \epsilon_i \sim N(0, \sigma)$

Smoothing helps when nearby nodes should have similar representations; over-smoothing removes differences that the task needs

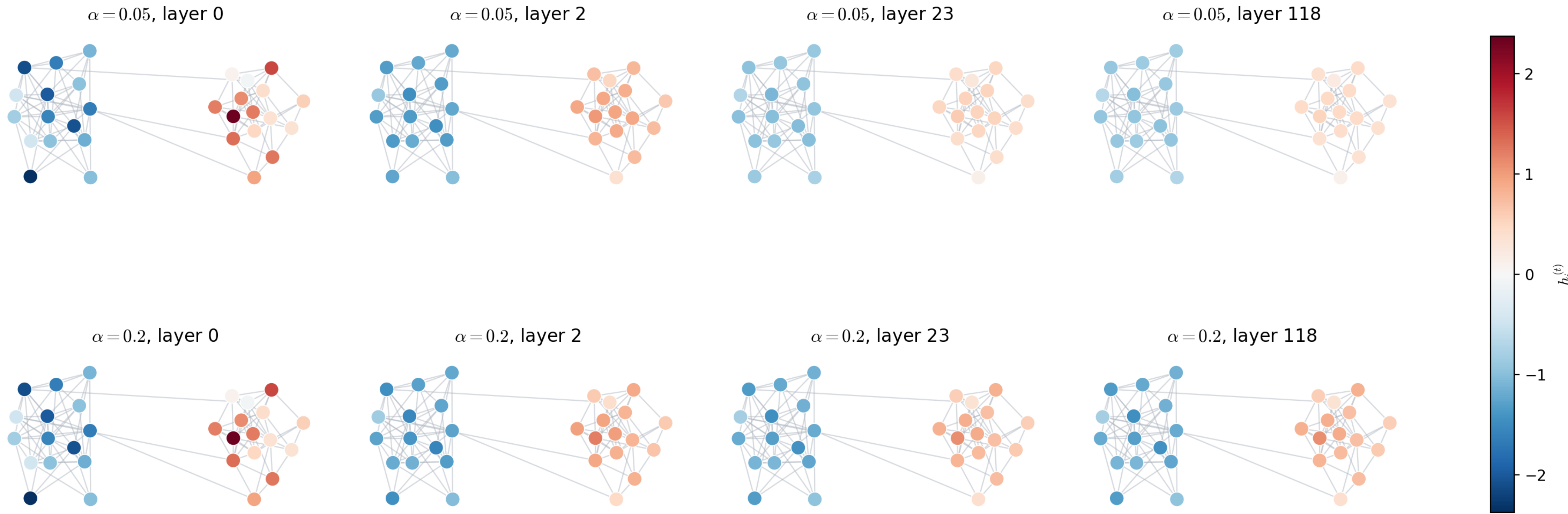


Over-smoothing

A possible solution

Repeatedly inject the initial features at each layer as residual connections:

$$H^{t+1} = (1 - \alpha)(I - \tilde{N})H^t + \alpha X, \quad \alpha \in (0, 1)$$



The tension between over-smoothing and over-squashing

We need many layers for information propagation (even if it is imperfect and subject to bottlenecks)

But too many layers make our representations collapse

How do we strike a balance?

Not a clear solution to these limitations in the literature...

Further thoughts on message-passing

Session 3 — Learning on Topological Data

Are GNNs learning the right things?

While GNNs have the ability to ignore the graph-structure in such cases, it is not clear that they will. In this work, we show that GNNs actually tend to overfit the graph-structure in the sense that they use it even when a better solution can be obtained by ignoring it

M. Bechler-Speicher, I. Amos, R. Gilad-Bachrach and A. Globerson, “Graph Neural Networks Use Graphs When They Shouldn’t”, *Proceedings of the 41st International Conference on Machine Learning* (2024).

Are the pairwise interactions even relevant?

More recently, there is a trend to decouple the input graph from the graph used for information propagation.

J. Topping, F. Di Giovanni, B. P. Chamberlain, X. Dong and M. M. Bronstein, "Understanding over-squashing and bottlenecks on graphs via curvature", *International Conference on Learning Representations*, (2022)

Are our datasets representative?

Our first finding is that standard benchmark graphs [...] cover only a small region of this graph space that GraphWorld is able to cover via synthetic graph generation.

J. Palowitch, A. Tsitsulin, B. Mayer and B. Perozzi, “GraphWorld: Fake Graphs Bring Real Insights for GNNs”, *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery & Data Mining* (KDD), 2022, pp. 3691–3701

Thank you for your attention!



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